

Inverse transformation:

(Using inverse transformation approach, we will generate random sample from continuous distribⁿ.)

Step ① Compute cdf of given distribⁿ ($F(x)$)

Step ② Set $F(x) = \underline{R}$ on the range of X — ~~⊗~~

Step ③ Solve the equatⁿ ⊗ for x in term R .
using ⊗, we have
 $x = \underline{F^{-1}(\underline{R})}$. — ~~⊗~~

Step ④ Generate uniform random sample $\sim U(0,1)$
 $R = \{ \underline{R_1}, \underline{R_2}, \dots, \underline{R_n} \}$. then using
⊗ generate a random sample.

$$R_1 \rightarrow x_1 = F^{-1}(R_1)$$

$$R_2 \rightarrow x_2 = F^{-1}(R_2)$$

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$$R_n \rightarrow x_n = F^{-1}(R_n)$$

$\Rightarrow \{x_1, x_2, \dots, x_n\}$
as random
sample from
given distribⁿ

$$F(x) = R$$

$$\rightarrow R \sim U(0,1)$$

$$\downarrow R \in (0,1)$$

$$R = 0.1, 0.2, 1, 0.5, 0.6$$

$R_i: 0.1, 0.2, 0.11, \dots$

$F(x)$ is cdf and $F(x) \in [0, 1]$

$$R \in [0, 1] \quad \text{and} \quad F(x) \in [0, 1]$$

$$F(x) = R$$

$$x = F^{-1}(R) \rightarrow \text{this deterministic}$$

$$R = \{0.1, 0.11, 0.6, 0.2, 0.9\} \in U(0, 1)$$

↓ Random sample from $U(0, 1)$

then generated sample will be

$$X = \{F^{-1}(0.1), F^{-1}(0.11), \dots, F^{-1}(0.9)\}$$

↓ random sample from given distribution

eg. ① Exponential distribution:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \lambda > 0$$

step ① We have to find out CDF of $f(x)$

$$F(x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^0 0 dx + \int_0^x \lambda e^{-\lambda x} dx$$

$$F(x) = 0 + \int_0^x \lambda e^{-\lambda x} dx$$

$$= \left[\frac{\lambda e^{-\lambda x}}{-\lambda} \right]_0^x = -e^{-\lambda x} + 1$$

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

Step (II) $F(x) = R \rightarrow F(x) \in [0, 1], R \in [0, 1]$

Step (III) $1 - e^{-\lambda x} = R \quad \text{--- (2)}$

$$+ e^{-\lambda x} = 1 - R$$

$$- \lambda x = \log(1 - R)$$

$$x = \frac{-1}{\lambda} \log(1 - R) \quad \text{--- (2)}$$

Step (IV) $R \in U(0, 1)$

$$\underline{R} = \{ \underline{0.1}, 0.21, 0.11, \dots, 0.9, 0.8 \}_{1 \times n}$$

Random sample from uniform (0, 1)

$$X = \{ x_1, x_2, \dots, x_n \}$$

Using eq. (2)

$$x_i = \frac{-1}{\lambda} \log(1 - R_i)$$

$$X = \left\{ -\frac{1}{\lambda} \log(1-0.1), -\frac{1}{\lambda} \log(1-0.2), \dots, -\frac{1}{\lambda} \log(1-0.8) \right\}$$

↓

this will be your generated sample from exponential distribⁿ.

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Random sample from uniform (a,b)

$$f(x) = \begin{cases} \frac{1}{b-a} & , x \in [a,b] \\ 0 & , x \notin [a,b] \end{cases}$$

Step ① CDF

$$F(x) = \int_a^x \frac{1}{b-a} dx$$

$$= \frac{x-a}{b-a}$$

$$F(x) = \begin{cases} \frac{x-a}{b-a} & , x \in [a,b] \\ 1 & , x \geq b \\ 0 & , x < a \end{cases}$$

Step ② $F(x) = R$

Step ③ $\frac{x-a}{b-a} = R \Rightarrow x-a = R(b-a)$

$$\boxed{x = a + R(b-a)}$$